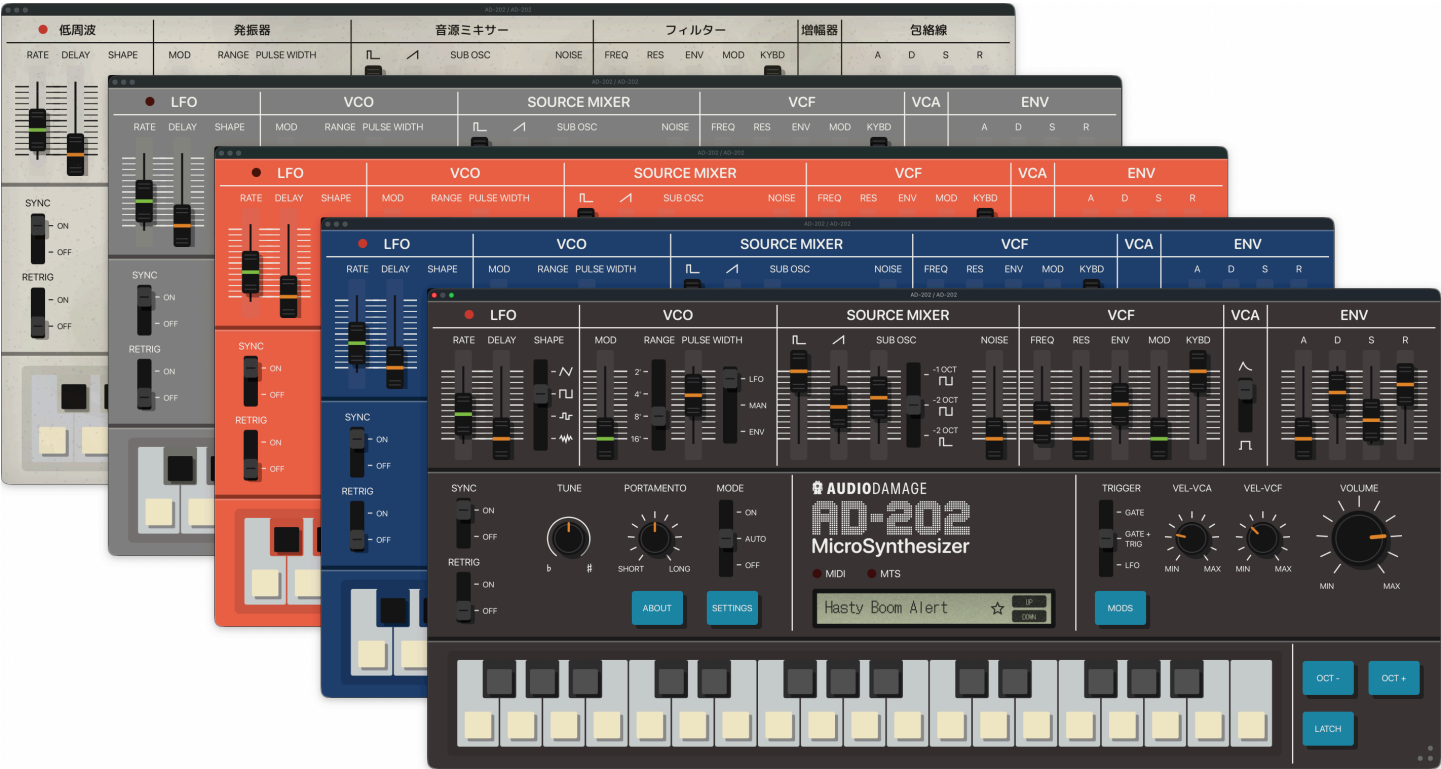


AD-202 Manual

Audio Damage, Inc.
Release 1.0



31 July 2026

© 2026 Audio Damage, Inc. All rights reserved.

The information in this document is subject to change without notice and does not represent a commitment on the part of Audio Damage, Inc. The software described by this document is subject to a License Agreement and may not be copied to other media except as specifically allowed in the License Agreement. No part of this publication may be copied, reproduced, or otherwise transmitted or recorded, for any purpose, without prior written permission by Audio Damage, Inc.

Credits

Software Design and Construction, Documentation

Chris Randall
Adam Schabtach

Field Testing

Stephen Boyd
Duncan Foster
Anders Göran
Chris Hahn
Steve Hamman
David Henkel
Matthew Lyon
varboska

Made Possible By

Elle
Tracie

License Agreement

BY INSTALLING THE SOFTWARE, YOU ARE CONSENTING TO BE BOUND BY THIS AGREEMENT. IF YOU DO NOT AGREE TO ALL OF THE TERMS OF THIS AGREEMENT, THEN RETURN THE PRODUCT TO THE PLACE OF PURCHASE FOR A FULL REFUND.

Single User License Grant: Audio Damage, Inc. ("Audio Damage") and its suppliers grant to Customer ("Customer") a nonexclusive and nontransferable license to use the Audio Damage software ("Software") in object code form solely on computers owned or leased by Customer.

Customer may make one (1) archival copy of the Software provided Customer affixes to such copy all copyright, confidentiality, and proprietary notices that appear on the original.

EXCEPT AS EXPRESSLY AUTHORIZED ABOVE, CUSTOMER SHALL NOT: COPY, IN WHOLE OR IN PART, SOFTWARE OR DOCUMENTATION; MODIFY THE SOFTWARE; REVERSE COMPILE OR REVERSE ASSEMBLE ALL OR ANY PORTION OF THE SOFTWARE; OR RENT, LEASE, DISTRIBUTE, SELL, OR CREATE DERIVATIVE WORKS OF THE SOFTWARE.

Customer agrees that aspects of the licensed materials, including the specific design and structure of individual programs, constitute trade secrets and/or copyrighted material of Audio Damage. Customer agrees not to disclose, provide, or otherwise make available such trade secrets or copyrighted material in any form to any third party without the prior written consent of Audio Damage. Customer agrees to implement reasonable security measures to protect such trade secrets and copyrighted material. Title to Software and documentation shall remain solely with Audio Damage.

LIMITED WARRANTY. Audio Damage warrants that for a period of ninety (90) days from the date of shipment from Audio Damage: (i) the media on which the Software is furnished will be free of defects in materials and workmanship under normal use; and (ii) the Software substantially conforms to its published specifications. Except for the foregoing, the Software is provided AS IS. This limited warranty extends only to Customer as the original licensee. Customer's exclusive remedy and the entire liability of Audio Damage and its suppliers under this limited warranty will be, at Audio Damage or its service center's option, repair, replacement, or refund of the Software if reported (or, upon request, returned) to the party supplying the Software to Customer. In no event does Audio Damage warrant that the Software is error free or that Customer will be able to operate the Software without problems or interruptions.

This warranty does not apply if the software (a) has been altered, except by Audio Damage, (b) has not been installed, operated, repaired, or maintained in accordance with instructions supplied by Audio Damage, (c) has been subjected to abnormal physical or electrical stress, misuse, negligence, or accident, or (d) is used in ultrahazardous activities.

DISCLAIMER. EXCEPT AS SPECIFIED IN THIS WARRANTY, ALL EXPRESS OR IMPLIED CONDITIONS, REPRESENTATIONS, AND WARRANTIES INCLUDING, WITHOUT LIMITATION, ANY IMPLIED WARRANTY OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, NONINFRINGEMENT OR ARISING FROM A COURSE OF DEALING, USAGE, OR TRADE PRACTICE, ARE HEREBY EXCLUDED TO THE EXTENT ALLOWED BY APPLICABLE LAW.

IN NO EVENT WILL AUDIO DAMAGE OR ITS SUPPLIERS BE LIABLE FOR ANY LOST REVENUE, PROFIT, OR DATA, OR FOR SPECIAL, INDIRECT, CONSEQUENTIAL, INCIDENTAL, OR PUNITIVE DAMAGES HOWEVER CAUSED AND REGARDLESS OF THE THEORY OF LIABILITY ARISING OUT OF THE USE OF OR INABILITY TO USE THE SOFTWARE EVEN IF AUDIO DAMAGE OR ITS SUPPLIERS HAVE BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES. In no event shall Audio Damage's or its suppliers' liability to Customer, whether in contract, tort (including negligence), or otherwise, exceed the price paid by Customer. The foregoing limitations shall apply even if the above-stated warranty fails of its essential purpose. SOME STATES DO NOT ALLOW LIMITATION OR EXCLUSION OF LIABILITY FOR CONSEQUENTIAL OR INCIDENTAL DAMAGES.

The above warranty DOES NOT apply to any beta software, any software made available for testing or demonstration purposes, any temporary software modules or any software for which Audio Damage does not receive a license fee. All such software products are provided AS IS without any warranty whatsoever.

This License is effective until terminated. Customer may terminate this License at any time by destroying all copies of Software including any documentation. This License will terminate immediately without notice from Audio Damage if Customer fails to comply with any provision of this License. Upon termination, Customer must destroy all copies of Software.

Software, including technical data, is subject to U.S. export control laws, including the U.S. Export Administration Act and its associated regulations, and may be subject to export or import regulations in other countries. Customer agrees to comply strictly with all such regulations and acknowledges that it has the responsibility to obtain licenses to export, re-export, or import Software.

This License shall be governed by and construed in accordance with the laws of the State of Colorado, United States of America, as if performed wholly within the state and without giving effect to the principles of conflict of law. If any portion hereof is found to be void or unenforceable, the remaining provisions of this License shall remain in full force and effect. This License constitutes the entire License between the parties with respect to the use of the Software.

System Requirements

The following table summarizes the operating system requirements and formats provided for AD-202. AD-202 is a 64-bit plugin.

Operating System	Minimum Version	Formats
macOS	10.13	AAX, AudioUnit, CLAP, VST3; Intel and Apple Silicon
Windows	10.0 version 1607	AAX, CLAP, VST3
Ubuntu	20.0	CLAP, LV2, VST3
iOS (separate purchase)	iOS 16	AUv3

Demonstration Version

We encourage you to download and try the demonstration version of AD-202 before purchasing it. The demo version of AD-202 is the same as the regular version, but has the following limitations:

- Presets cannot be saved, nor can parameter values or other settings. This includes the information usually stored by your host DAW. If you save a DAW session with an instance of the demo version of AD-202, the plugin will revert to its default state when you reload the session.
- AD-202 will cease to emit audio altogether 20 minutes after you add it to your DAW session. You can remove it and add it again, but it will revert to its default state.

If you purchase AD-202 after using the demonstration version, simply run the installer provided to you after your purchase to replace the demo version with the full version.

Introduction

AD-202 is a monophonic synthesizer with one oscillator, one filter, one envelope, and one LFO.

That sentence is either a warning or an invitation, depending on your temperament. If you're used to instruments with six oscillators and a modulation matrix, a single-oscillator monosynth can look like a toy. It isn't. Some of the most recognizable sounds in the last forty years of recorded music came out of instruments built exactly this way, and there is a reason for that: when there is only one of everything, every control matters, and you can hear what each one does. You stop configuring and start playing.

When we conceived of this plugin, the instrument we had in mind was the MC-202, and the SH-101 whose voice circuit it borrowed. Those two share a design that is close to ideal for its size. A single oscillator produces a sawtooth and a pulse whose width you can narrow by hand or by modulation. A sub-oscillator, formed by a simple divider hung off the main oscillator and always perfectly in tune, makes these instruments punch far above their weight in the bass. A noise source rounds out a four-input mixer that deliberately clips on its way into a four-pole ladder lowpass filter. One envelope opens the filter and shapes the amplitude at the same time. One LFO wobbles whatever you point it at. That's the whole instrument, and it is enough.

We built AD-202's voice from both hands-on experience and service-manual research, and kept the parts of the original's behavior that a more "correct" design would have quietly improved away. The pulse width only narrows, from square toward a sliver, because that is the only direction the original circuit could go. Pulse-width modulation from the LFO always uses the triangle wave no matter what the LFO shape switch is set to, because that is how the circuit was wired. The filter gets *louder* as you add resonance rather than thinner, which is a side effect of how the hardware's output stage was arranged and is a large part of why these instruments sound aggressive rather than polite when you open them up. The mixer clips. If any of these design decisions were accidental, history has shown that none of them are defects.

Once we'd nailed the voice, we went further to build a software plugin with modern conveniences not available to hardware builders of last century. The original was a monosynth attached to a sequencer, and the sequencer was most of the point of it in 1983. Today we're all using far more powerful and convenient DAWs and other software tools, so there was no reason to put a redundant step sequencer in a plugin. AD-202 will run as an eight-voice polyphonic instrument when you want chords, with a stereo spread control to place those voices across the field. An Age control gives each voice its own permanent set of small mis-calibrations, the way an old instrument's voice circuits drift apart from one another over the decades. This detuning is pleasant in mono and something close to a chorus in poly. Three separate saturation stages let you dirty the signal before the filter, inside the filter, or after the amplifier, each with its own character. There is a tilt EQ at the end to place the result in a mix. The LFO will lock to your host's tempo. Velocity and aftertouch, neither of which the original had any notion of, route to the places where they do the

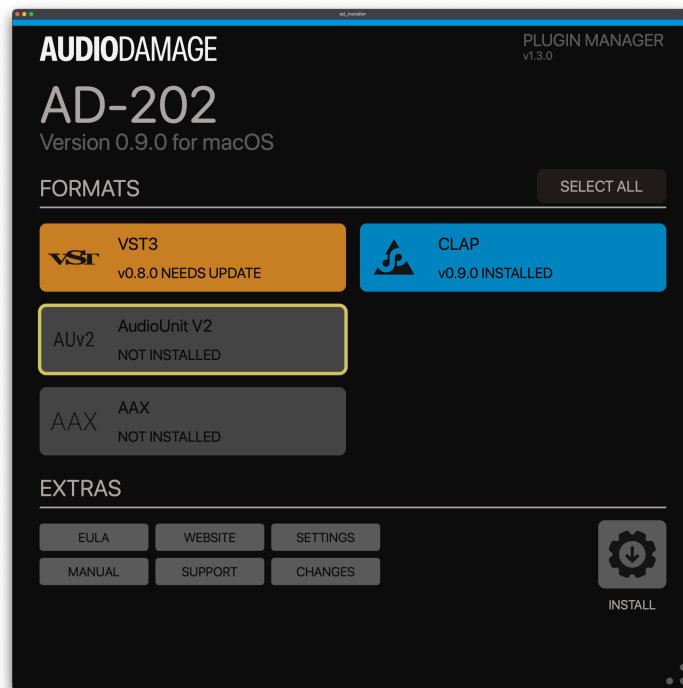
most good. Finally, the whole thing speaks MPE and MTS-ESP for those who want per-note expression or tunings other than the twelve we usually agreed on.

The panel reproduces the MC-202's, down to the sampled colors, the lever switches, and the chicklet keys. Five finishes are included. Four of them are colors. The fifth is not, quite, and if you recognize the name we don't need to explain it to you.

This manual gives an overview of AD-202 followed by detailed descriptions of all of its controls. You are welcome to install it, flip through the presets, and start pulling faders until something sounds good. With an instrument this size that is a genuinely efficient way to learn it, and we'd be a little disappointed if you didn't. But there is more here than the front panel admits, particularly in the MODS panel, and it's worth coming back to read once you've had a chance to play. We hope you enjoy using it as much as we enjoyed building it.

Installation

AD-202 uses our custom plugin manager application for installation. Launch it as usual on your operating system of choice and you'll be presented with a window like this:



Near the top of the window, beneath the name of the plugin, you'll see the version number of the software carried by the installer. This is distinct from the version of the plugin manager itself, which is shown in the upper right and usually not of much interest. (Note that the version numbers in this screenshot are for demonstration purposes only and may not reflect the actual version numbers of the software at the time you read this manual.)

Under the heading FORMATS are large buttons corresponding to the plugin formats which can be installed: AAX, AU, CLAP, LV2 and/or VST3, depending on the operating system. If the plugin is already present on your system in one or more formats (i.e. if you're upgrading from a previous version), the corresponding button is drawn in blue. When possible, the version number of the existing plugin is also shown. If the installer contains a newer version of the plugin than the one(s) already installed, the button is drawn in orange to alert you to the fact that an update is available. If a format is not present on your system, its button is drawn in grey.

Click a button to select the format for installation. A yellow outline appears around the button to indicate that its format will be installed. In the above screenshot, AAX and AudioUnit are not installed, VST3 is installed but older than the version contained in the installer, CLAP is installed and up to date, and the AudioUnit is selected for installation. Clicking a button a second time removes the yellow outline, and the corresponding format will not be installed. Clicking the SELECT ALL button selects all available formats for installation.

No changes to your system's storage device take place until you click the INSTALL button near the lower-right corner of the window. Click that button and you'll receive visual confirmation that the format(s) you've selected have been installed. (Yes, it happens quickly. Our products are not encumbered by any DRM or anti-piracy baggage and hence install more quickly than many others.) On Windows and Linux, if you hold down the Shift key on your keyboard, the INSTALL button's label switches to

UNINSTALL, and clicking it will remove the selected formats from your system¹. Once you're installed and/or removed the formats you need, simply close the application in the usual manner for your operating system. You're done. There is no license code or other authorization necessary; we'd rather assume we can trust you than burden you with an onerous DRM system.

You'll find some handy buttons under the EXTRAS heading, all of which are pretty self-explanatory:

- EULA: presents the End-User License Agreement for our products. By clicking the INSTALL button you're implicitly agreeing to these terms, but we expect that you'll find them reasonable should you take the time to read them.
- MANUAL: opens the current version of this user manual, in PDF form, in your web browser.
- WEBSITE: opens the product's web page in your browser.
- SUPPORT: displays information for contacting us, either via our Discord presence or through email.
- SETTINGS: produces a panel in which you can choose the installation locations for each plugin format. Note that these locations are nominally stipulated by the publishers of the formats and changing them might cause problems for some hosts.
- CHANGES: displays a summary of what we've added, changed or improved since the last version of the product. This button isn't present on the first release of the product.

¹ Blame Apple, not us, for the lack of this feature on macOS. On macOS just manually delete the plugin(s) from your plugin folder(s).

Operation

AD-202 is a monophonic (and, when you want it to be, polyphonic) subtractive synthesizer built around a single oscillator, a four-pole ladder lowpass filter, one shared envelope, and one LFO. This small feature set is the reason the instrument is easy to play: there is exactly one of everything, and every control does one obvious thing.



The main panel presents the synthesis controls in the six sections printed across the top half of the window: LFO, VCO, SOURCE MIXER, VCF, VCA, and ENV. The lower half of the panel holds the performance controls (tuning, portamento, trigger behavior, velocity response, and output level) along with the preset display and a couple of status indicators. Below the panel is a keyboard strip you can play with the mouse or your finger on touchscreen devices.

Two additional panels slide over the main window. MODS holds a set of secondary sound parameters that are saved with the preset: voice mode, the saturation stages, and the aftertouch routings. SETTINGS holds global preferences that are not saved with the preset. Both are described below.

VCO

The oscillator section produces a sawtooth and a variable-width pulse from a single master oscillator, plus a sub-oscillator derived from that oscillator by a flip-flop divider chain. The saw and pulse are anti-aliased so they stay clean at the top of the keyboard.

RANGE is the octave switch, setting the footage of the oscillator: 16', 8', 4', or 2'. The default is 8', which puts middle C where you expect it to sound.

PULSE WIDTH sets the width of the pulse wave, or the depth of its modulation, depending on where the switch beside it is set. The control only narrows the wave: at minimum the pulse is a 50% square, and turning it up narrows the wave toward roughly 2%. This matches the behavior of the original hardware, where the width control could only take the wave away from square. Since a 50% square has no even harmonics, the fader's effect is dramatic near the bottom of its travel, changing the sound from hollow to nasal almost immediately, and then progressively thinner as you keep moving the slider upwards.

The three-position switch beside the Pulse Width fader chooses what controls the width:

- MAN: the fader sets the width directly, and it stays there.
- LFO: the LFO sweeps the width, with the fader setting the depth of the sweep. Note that the LFO source here is *always the triangle wave*, regardless of what the SHAPE switch in the LFO section is set to. This is how the original circuit was wired, and it is what makes the classic PWM string pad work no matter what else the LFO is doing.
- ENV: the envelope sweeps the width, with the fader setting the depth. Higher envelope output means a narrower pulse, so the wave thins out during the attack and fills back in during the release.

MOD sets the depth of LFO modulation of the oscillator's pitch: vibrato, in other words. At minimum there is none. At maximum the LFO swings the pitch across a full octave, which is well past vibrato and into siren territory. Unlike the pulse-width modulation, this one uses whichever waveform the LFO SHAPE switch has selected, so with the square wave you get a trill and with the random waveform you get a sample-and-hold pitch sequence.

Source Mixer

The Source Mixer combines the four sound sources and feeds them to the filter. The four faders are, from left to right, pulse, sawtooth, sub-oscillator, and noise. The first two are marked with waveform glyphs; the second two are marked SUB OSC and NOISE.

There is no gain staging to worry about beyond your own taste, but note that the mixer soft-clips on its way into the filter. Running all four faders up is not a mistake; it is a tone color, and it is where a lot of the instrument's grit comes from. If you want more of that grit, the PRE-FILTER SAT control in the MODS panel pushes the same clipper harder.

The pulse, sawtooth, and noise faders operate as you'd expect, setting the levels of the corresponding signal. The SUB OSC sets the level of the sub-oscillator, and the three-position switch beside it chooses which of three sub waveforms you get:

- -1 OCT square: a square wave one octave below the oscillator.
- -2 OCT square: a square wave two octaves below.
- -2 OCT pulse: a narrow pulse two octaves below, produced by combining the two divider stages. This gives a 75/25 duty cycle, which is thinner and buzzier than the square and sits better under a bass line without turning it to mud.

The sub is generated by a divider running off the master oscillator, so it is always exactly in tune with it. It is gently filtered above 6kHz and DC-blocked.

NOISE sets the level of the white noise generator. The noise is rolled off above 16kHz, which takes the hiss off the very top and matches the character of the original. Noise is useful in small amounts as a texture inside a filtered patch, and in large amounts as a source in its own right for wind and percussion sounds.

VCF

The filter is a four-pole (24dB/octave) resonant ladder lowpass. Five faders control it.

FREQ sets the cutoff frequency, from 10Hz to 20kHz. The taper is skewed so that the middle of the fader's travel lands around 1kHz, spreading the the useful range across more of the fader range. This slider sets the base value of the filter frequency; the modulation sources described below add to this setting.

RES sets the resonance, from none at the bottom to self-oscillation at the top of the travel. As on the original, the filter's output level rises with resonance rather than falling, which is a consequence of how the hardware's phase splitter was arranged. The practical effect is that turning up the resonance makes the patch louder and more aggressive rather than thinner, so you may find yourself pulling the Volume knob back as you go.

ENV sets how much the envelope opens the filter. The modulation is positive-only and spans up to nine octaves at full, which is the entire range of the filter. It may be the single most expressive control in the instrument: it is what turns a static tone into a plucked, swept, or wobbling one. Note that the envelope driving it is the same envelope driving the VCA, so the filter sweep and the amplitude contour are always locked together.

MOD sets how much the LFO modulates the cutoff. This one is bipolar around the current cutoff setting, spanning up to plus or minus four octaves at full. With the triangle waveform this gives you the classic filter wobble; with the square it gives you a two-step alternation; with the random and noise waveforms it gives you burbling and hiss respectively.

KYBD sets the keyboard tracking of the cutoff, that is, how much the note you play moves the filter along with it. At 100% (the default) the cutoff tracks the keyboard one-for-one, so the timbre stays consistent as you move up and down the keyboard. At 0% the cutoff stays where you put it regardless of pitch, which means high notes get progressively duller and low notes progressively brighter. Tracking is referenced to middle C, so that note sounds the same at any Kybd setting.

The filter also has a drive control, but it lives in the MODS panel with the other saturation stages. See PRE-FILTER SAT and FILTER DRIVE below.

VCA

The VCA section is a single two-position switch which selects what controls the amplifier:

- ENV (upper position, marked with an envelope glyph) — the shared envelope controls the amplitude, so the sound follows the ADSR contour.
- GATE (lower position, marked with a gate glyph): the amplifier follows the key gate directly, snapping on when a key goes down and off when it comes up. There is a two-millisecond smoothing lag on the gate so it doesn't click.

Gate mode does not disable the envelope; it just takes the envelope out of the amplitude path. The envelope keeps running and keeps modulating the filter, so a common trick is to set Gate mode and use the envelope purely as a filter contour, giving you an organ-like amplitude with a moving timbre.

ENV

One envelope generator serves the filter, the amplifier, and optionally the pulse width. It is a conventional ADSR with the segment curvature of an analog envelope rather than the straight lines of a digital one.

- A (Attack) sets the time from key-down to full level, from 1.5ms to 4 seconds.
- D (Decay) sets the time from full level down to the sustain level, from 2ms to 10 seconds.
- S (Sustain) sets the level held while the key is down, from silence to full.
- R (Release) sets the time from key-up back to silence, from 2ms to 10 seconds.

The Attack, Decay, and Release faders are skewed so that the short times occupy a generous portion of the travel, since that is where percussive sounds live. Retriggering the envelope re-enters the attack segment *from the current level* rather than from zero, which is what an analog envelope does and what makes fast repeated notes sound continuous instead of chopped. Whether a given note retriggers the envelope at all depends on the TRIGGER switch; see below.

LFO

The LFO runs continuously, whether or not a note is held, and the red LED at the top of the section blinks along with it.

RATE sets the LFO speed, from 0.1Hz to 30Hz. If the SYNC switch below is set to ON, the fader instead selects a metrical division of the host tempo, from two whole bars down to a thirty-second note, with dotted and triplet values in between.

DELAY sets a fade-in time for the LFO, from zero to 4 seconds. With Delay up, the LFO's depth ramps up from nothing over the delay time after each note starts, rather than being applied immediately. This is the standard way of getting delayed vibrato, where a note starts steady and then begins to waver. The delay affects the LFO's *depth* only, not its phase, so the LFO keeps running underneath and the trigger behavior described below is unaffected. The ramp restarts on each new note attack.

SHAPE selects the LFO waveform from four positions, top to bottom:

- Triangle: a symmetrical rise and fall. The default, and what you want for vibrato and filter wobble.
- Square: an instant alternation between two values.
- Stepped random: a new random value at the start of each LFO cycle, quantized to a coarse staircase the way the original's six-bit converter was. The stepping is audible and is part of the character.
- Noise: a continuously changing random value updated at control rate, which is much faster and rougher than the stepped random.

SYNC locks the LFO rate to the host tempo, changing the Rate fader from a frequency control to a note-division control as described above.

RETRIG restarts the LFO's cycle from the beginning at each new note. With Retrigger off the LFO free-runs, so a given note starts wherever the LFO happened to be, and repeated notes get slightly different modulation each time. With it on, every note gets identical modulation, which is what you want when the LFO is doing something rhythmic.

Performance Controls

These sit across the lower half of the panel.



TUNE is the master tuning control, plus or minus 50 cents, marked with flat and sharp symbols at the ends of its travel. This tunes the whole instrument, not individual voices.

PORTAMENTO sets the glide time between notes, from zero to five seconds, marked SHORT and LONG. The glide is exponential in pitch, meaning it moves at a constant rate in octaves per second, so a glide across an octave takes the same time no matter where on the keyboard it happens.

MODE, the switch beside the Portamento knob, sets when the glide applies:

- ON: every note glides from the previous one.
- AUTO: notes glide only when played legato, that is, when a new key goes down while the previous one is still held. Staccato notes jump straight to pitch. This is the default and it is the most musically useful of the three, since it lets you choose note by note whether to glide.
- OFF: no glide.

TRIGGER sets the keyboard priority and the retrigger behavior together, which are two aspects of the same circuit on the original hardware:

- GATE: low-note priority with single triggering. The lowest key held sounds, and the envelope fires only when you go from no keys held to one. Playing legato moves the pitch without restarting the envelope.
- GATE + TRIG: last-note priority with multiple triggering. The most recently pressed key sounds, and every new key restarts the envelope. This is the default, and it is the behavior most people expect from a keyboard.
- LFO: low-note priority, with the envelope retriggering on every cycle of the LFO for as long as a key is held. This turns the LFO into a repeat clock and the instrument into a self-playing arpeggiator of one note. Set a short envelope, turn the Rate up, and hold a key.

VEL-VCA sets how much MIDI velocity controls loudness. At minimum, every note plays at full level regardless of how hard you hit it, which is how the original behaved (it had no velocity sensing at all). Turning it up makes soft notes quieter.

VEL-VCF sets how much velocity controls the filter cutoff. As with Vel-VCA, the effect is subtractive: at higher settings, soft notes pull the cutoff down, by as much as four octaves at maximum. Notes played at full velocity always sound at the cutoff you dialed in, so this control never brightens a patch beyond where you set it; it only darkens the quiet notes.

VOLUME is the output level, running from silence to +6dB, marked MIN and MAX. It sits at the very end of the signal path, after the distortion and tilt EQ stages in the MODS panel.

Keyboard Strip

Below the panel is a 32-key keyboard running from F to C, the same span as the original's, which you can play with the mouse or your finger on a touchscreen. It is a convenience for auditioning patches, not a performance instrument: the on-screen keys have no velocity sensitivity and always play at a fixed medium velocity. If you want to hear what the Vel-VCA and Vel-VCF controls do, play the plugin from a MIDI keyboard.

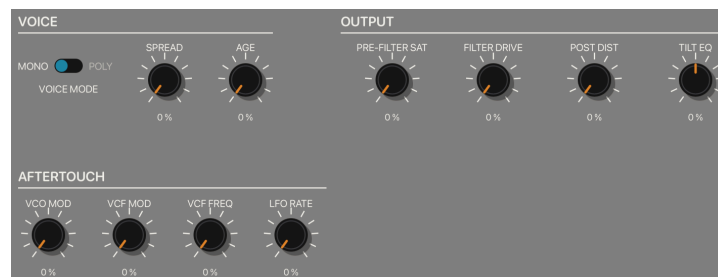
OCT + and OCT - shift the keyboard up and down by one octave at a time, up to two octaves in either direction from the default position.

LATCH holds the keys you press until you press them again, so you can leave a note or a chord sounding while you work on the patch with both hands free. Turning Latch off releases everything currently held. Latch applies to incoming MIDI as well as to the on-screen keys.

Two indicators sit above the keyboard, beside the preset display. MIDI lights when the plugin receives MIDI input, which is the first thing to check when a patch is silent. MTS lights when an MTS-ESP tuning master is running, in which case AD-202 takes its tuning from that master rather than from standard equal temperament, and notes that the master leaves unmapped will not sound.

MODS Panel

Clicking MODS in the upper right slides in a panel of secondary sound parameters. Everything here is a real parameter and is saved with the preset, unlike the contents of the Settings panel. Click CLOSE to dismiss it.



Voice

VOICE MODE switches between MONO and POLY. In Mono the instrument behaves like the hardware it is modeled on: one voice, with the keyboard priority and trigger rules described above under TRIGGER. In Poly you get eight voices, allocated to the oldest

free voice and, when all eight are busy, stealing the oldest sounding one. Portamento in Auto mode applies in Poly only when a voice is reused while still sounding.

SPREAD pans the voices apart in the stereo field, alternating left and right of center. It has no effect in Mono, where the output is always centered. At minimum all voices are centered; at maximum they are spread to the edges. The panning is equal-power, so the overall level stays put as you open the spread.

AGE applies a fixed random detuning and mis-calibration to each voice, in the manner of an old instrument whose voice circuits have drifted apart. Each voice gets its own permanent set of offsets, applied to pitch (up to about 20 cents), pulse width, filter cutoff (up to a quarter octave), LFO rate, and the four mixer levels. In Mono this reads as a slightly-off instrument; in Poly it reads as a chorus, since the voices are now genuinely different from each other. Small amounts do a lot.

Output

These four controls are the saturation and tone-shaping stages, arranged in signal order.

PRE-FILTER SAT drives the source mixer harder into its soft clipper, ahead of the filter. At minimum the clipper is at its stock level and only clips when you have several mixer faders up. Turning it up pushes the mix into the clipper by as much as a factor of eight, which thickens the sound and adds harmonics *before* the filter gets to remove them. This is the most musical of the three distortion stages because the filter tames whatever it produces.

FILTER DRIVE adds saturation inside the filter's feedback path. This is a different flavor from Pre-Filter Sat: rather than dirtying the input, it changes how the filter itself behaves under load, softening the resonance and giving the whole filter a rounder, more compressed character. The drive costs level as you turn it up, so a makeup gain rides along behind it and keeps the loudness roughly constant.

POST DIST is a distortion stage after the VCA, at the very end of the voice path. Because it is after the amplifier, it distorts the envelope contour as well as the waveform, flattening the dynamics and making short notes sound louder than they are. It is the bluntest of the three, and it is the one to reach for when you want the patch to sound broken. It too has a makeup gain, so turning it up does not simply make things louder.

TILT EQ is a bipolar tilt equalizer applied after the distortion. Turning it left removes highs and adds lows; turning it right does the opposite. Straight up is flat. It is a fast way to place a patch in a mix without reaching for a separate EQ.

Aftertouch

The four knobs in this section route MIDI channel pressure (or, in MPE mode, per-note pressure) to four destinations. All four are additive: they add to whatever the corresponding panel control is already doing, so a patch with no vibrato at rest can have vibrato on demand.

VCO MOD adds LFO-to-pitch depth with pressure, which is the standard way to get vibrato under your fingers rather than permanently on.

VCF MOD adds LFO-to-filter-cutoff depth with pressure.

VCF FREQ opens the filter with pressure, by as much as four octaves at full. This is the most immediately expressive of the four: lean into a note and it brightens.

LFO RATE speeds up the LFO with pressure, by as much as a factor of four. Combined with VCO MOD, this gives a vibrato that gets both deeper and faster as you press, which is how a string player actually plays.

If your controller sends stepped or noisy pressure, the Pressure Smoother in the Settings panel will smooth it; see below.

Settings Panel

Clicking SETTINGS opens the global preferences. Unlike the MODS panel, nothing here is saved with the preset. Changes apply to the current instance immediately; click SAVE to make them the defaults for new instances. Closing the panel without saving leaves the changes in effect for this instance only.

SHOW TOOLTIPS turns the hover tooltips on and off.

PANEL COLOR selects the color of the panel. We started with the colors of the cases of the MC-202 and SH-101, added a contemporary black and another derived from nostalgia for a consensual hallucination.

MPE MODE enables MPE, in which each note arrives on its own MIDI channel and carries its own pitch bend and pressure. With MPE off, the plugin runs in the conventional single-channel mode where bend and pressure apply to everything sounding. Turn this on only if your controller actually sends MPE; turning it on otherwise will make bends behave strangely.

PITCH BEND OVERRIDE, and the PB RANGE knob beside it, let you set the pitch bend range yourself instead of accepting the range the controller declares. With Override off, the controller's declared range is used. With it on, the PB Range knob sets the range in semitones. This is mainly useful with MPE controllers, which typically declare 48 semitones, when you want something narrower.

PRESSURE SMOOTHER, and the two knobs beside it, smooth incoming aftertouch. MIDI channel pressure is only seven bits, and many controllers send it in visible steps; the smoother turns those steps into a continuous ramp. The two knobs set the rise and fall times independently, so you can have pressure that responds quickly but decays gently.

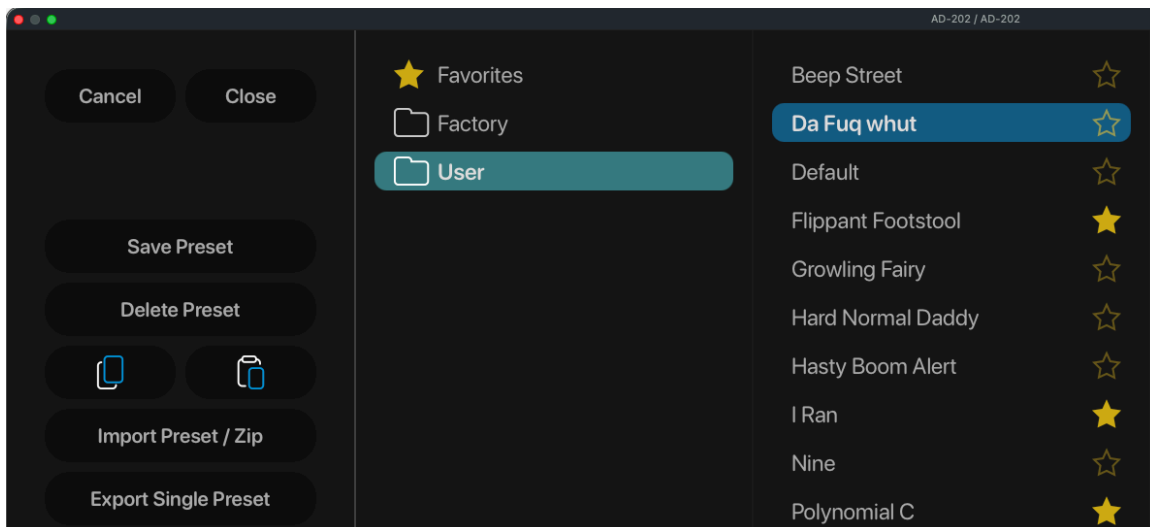


Presets

AD-202 includes a collection of presets to serve as a demonstration of its capabilities and inspirations for your own creations. There are a few controls near the center of the window associated with presets:



The name of the current preset appears in the center. (You probably figured that out yourself.) Clicking the UP and DOWN buttons loads presets in alphabetical order. Clicking the star outline marks the preset as a favorite to help you find it again in the future. To examine all of the presets, click the name of the current to open the preset browser.



The browser displays presets and folders in scrollable lists, arranged in columns. The leftmost list shows the folders within AD-202's preset collection, grouped in two categories: Factory and User. Clicking any of these folders reveals its contents in the next list. Clicking on a preset name loads the settings into AD-202. Click the CLOSE button in the preset browser to dismiss it. If you

click the CANCEL button instead, the browser closes and AD-202's settings revert to their previous state. Double-clicking a preset name loads the preset and dismisses the preset browser. Loading a preset irretrievably erases the current settings, so if you have created something that you want to use again, save it as a new preset before loading another preset.

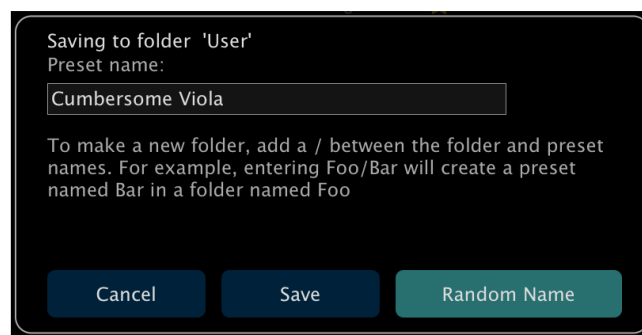
Once you have clicked on any item in the panel, you can navigate within the preset browser with the keys on your keyboard. The left and right arrow keys move the selection between columns, and the up and down arrow keys move it within the list. Tapping the ESC key has the same effect as clicking the Cancel button.

Just like the star at the top of the window which we mentioned previously, the stars to the right of preset names mark presets as favorites. Clicking a filled star removes the mark from a favorite. Once you've marked at least one favorite, a correspondingly named category appears in the leftmost column. Clicking it shows you all the presets you've marked, and clicking their names loads them as usual.

The folders and presets in the browser correspond to folders and files within AD-202's own folder on your storage device (i.e. your computer's hard drive or SSD). This folder is located at `C:\ProgramData\Audio Damage\AD202\` on Windows, and `~/Music/Audio Damage/AD202/` on macOS and Linux. You can store your presets anywhere you like, but for them to show up in the User list they must be placed in the User folder within AD-202's folder. Also, to avoid possible collisions during future updates, do not store your presets within the Factory folder.

Any folders you create within the User folder will show up as folders in the User list. You can create sub-folders within the User folder, and sub-folders within those folders. You can't nest folders deeper than that because the preset browser has only four lists.

To save your presets, click the SAVE PRESET button at the left edge of the window. This invokes a dialog box with a couple of helpful features. As the text therein describes, you can create a folder within the destination folder (whose name is given at the top of the dialog box) by adding the folder's name to the beginning of the preset's name, separated by a slash mark.



Clicking the RANDOM NAME button replaces the preset's name with a pair of words chosen at random from two lists. While the resulting names won't have any connection with what the plugin is doing, you may find this feature useful for coming up with alternatives to routine names like "My Preset 12".

Potential pitfall: once you've saved a preset, clicking its name in the list loads the preset, overwriting whatever changes you've made since you saved the preset. Hence if you want to save the preset again to preserve the changes you've made, do not click on its name before saving it.

You can delete presets and folders from the lists by clicking their name and then clicking the DELETE PRESET or DELETE FOLDER button. AD-202 will give you a chance to confirm this action or cancel it. If you confirm, the preset/folder will be removed from your storage system and is gone for good.

Importing and Exporting Presets

Preset files are plain-text XML files so that you can exchange them online in forums, copy them between a Windows computer and a Macintosh (and even between an iPad and a regular computer), email them to your friends, etc.

The two buttons with icons representing copying and pasting (copy on the left, paste on the right) copy AD-202's current settings to the system clipboard and paste settings from the clipboard. You can use the copy and paste commands to transfer settings between two instances of AD-202 or paste the settings into an email message or text editor. When copied to the clipboard, presets are presented in the same XML text as used in preset files.

The IMPORT PRESET/ZIP button provides a way to add presets to AD-202 without manually moving them into the appropriate folders in your file system. Clicking this button produces a file-browser window wherein you can select either a single preset file or

a .zip file containing one or more presets. After you select the file, AD-202 copies the preset(s) into whichever folder you have selected in AD-202's preset list, unzipping the file first if necessary.

Depending on whether you've selected a preset or folder, the EXPORT SINGLE PRESET or EXPORT FOLDER AS ZIP button performs the complementary functions of the Import button. First select either a preset or a folder in AD-202's list, then click the export button. A file-save window appears; choose a location in your file system, give the file a name, and click SAVE. If you have chosen a folder in AD-202's preset list, the plugin places it and all of the presets it contains in a .zip file.

Default Preset

If you save a preset with the special name "Default" in the User folder, new instances of AD-202 will load it automatically when you add it to your DAW session. You can use a default preset file to give yourself the same starting point with AD-202 whenever you use it. The plugin installer creates a default preset file for you but feel free to replace it with your own.

And Finally...

Thanks for purchasing AD-202. We make every effort to ensure your satisfaction with our products and want you to be happy with your purchase. Please write to support@audiodamage.com if you have any questions or comments.

Document Revisions

- 2026-07-28: Initial draft for version 1.0